

S.M. Abdullah

Full-Stack Software Developer • FastAPI · React · Next.js · PostgreSQL
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JKKIH, Jahangirnagar University, Savar

PROFESSIONAL SUMMARY

Backend focused full-stack software developer with hands-on experience building modern web applications, AI-assisted platforms, and real-time systems. Proficient in FastAPI, React, PostgreSQL, and vector databases, with expertise in REST API development, authentication, embeddings, semantic search, and cloud deployment. Adept at designing reliable, scalable solutions and continuously exploring new technologies to create meaningful user experiences. Currently a third-year IIT student at Jahangirnagar University, open to opportunities where I can own features end-to-end and work alongside engineers who push back.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C++, HTML, CSS

Frameworks: FastAPI, React, Next.js (learning), Django

Databases: PostgreSQL, MySQL, SQLite, Supabase, Pinecone (vector DB)

Tools & Platforms: Git/GitHub, Docker, Postman, Thunderclient, REST API design, pgAdmin, Crawlee, Playwright, Selenium.

Specialties: API development, authentication flows, RAG pipelines, web crawling & scraping (Crawlee, Playwright, Selenium), LangChain, embeddings, semantic search, cloud deployment (Netlify)

Soft Skills: Strong problem-solving · Critical thinking · Quick learner · Adaptability · Team collaboration

EXPERIENCE

Full-Stack Developer - Gudam (ICT Division Project)

2024 – Present

- Built an ML microservice for behavioral commerce intelligence: session intent scoring, RFM-based churn prediction, user behavioral vectors, and hybrid semantic search with intent-aware reranking.
- Integrated a RAG-powered AI commerce assistant using Pinecone as the vector store and Hugging Face embeddings for semantic retrieval.
- Designed the full FastAPI service architecture from schema to deployment; wrote tests throughout.

PROJECTS

BehaviorIQ – AI-Powered Behavioral Commerce Intelligence Platform

FastAPI + Python + Pinecone + PostgreSQL | [Github](#)

- E-commerce platforms treat every visitor the same regardless of behavior, built a FastAPI microservice that scores live session intent (0-100) from browsing signals and predicts RFM-based churn probability, turning raw pixel events into an actionable behavioral profile for every visitor
- Static product rankings ignore who's actually searching, built user behavioral vectors and a hybrid search pipeline that combines vector similarity with intent-aware reranking, so results shift based on each visitor's real-time intent score instead of a one-size-fits-all ranking

- SME owners had no way to ask questions about their own commerce data, built a RAG-powered AI assistant on top of the behavioral data layer, letting owners query intent trends, churn risk, and product performance in plain English instead of digging through dashboards

Ember - AI Mental Health Companion *React · FastAPI · Supabase · Hugging Face*

- Full-stack AI mental health support platform integrating Hugging Face toxicity detection models with a React frontend and FastAPI backend.
- Supabase handles persistent user data and authentication

Live: embermentalhealthcompanion.netlify.app

Onkur - AI Plant-Care App *React · FastAPI · AI/ML*

- Personal plant-care application with AI-driven daily care routines, weather-based recommendations, and image-based plant health diagnostics via Plant.id API.

Live: onkur.netlify.app

Universal Web Scraper & RAG Chatbot *Crawlee · Playwright · LangChain · Gemini · Pinecone*

- Built a production-grade web crawler using Crawlee + Playwright for dynamic site traversal and Trafilatura for clean content extraction.
- Implemented a full RAG pipeline: LangChain chunking → sentence-transformers embeddings → Pinecone vector store → Gemini LLM generation.
- System handles arbitrary URLs and lets users query any crawled site through a natural language interface.

Personal Blog Platform *FastAPI · Supabase · REST API*

- Blog platform with full CRUD, user authentication, and a responsive UI. Designed and implemented the REST API and data layer end-to-end.

EDUCATION

B.Sc. in Information and Communication Technology

Institute of Information Technology, Jahangirnagar University

Third Year · 5th Semester | **CGPA: 3.82**

ACHIEVEMENTS & ACTIVITIES

- Prompt Engineering Competition by AgentX — 8th Place